Lecture 18

Operators, Schrodinger Equation

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Yesterday we saw how to extract information from the wavefunction. In particular, we saw that we could calculate average values of observables.

Position average:

$$\langle x \rangle = \int_{-\infty}^{\infty} dx\, \psi^*(x,t)\, x\, \psi(x,t)$$

Other observables like functions of position, for example, if potential energy is

$$V = \frac{1}{2}kx^2$$

Average potential energy:

$$\langle V \rangle = \int_{-\infty}^{\infty} dx\, \psi^*(x,t)\, \frac{1}{2}kx^2\, \psi(x,t)$$

There are other observables which cannot be written in terms of x. For example, momentum: p.

There is no function $p(x)$ because of the uncertainty principle. We want to find something that can be put in the integral above like:

$$\langle p \rangle = \int_{-\infty}^{\infty} dx\, \psi^*(x,t)\, \hat{p}\, \psi(x,t)$$

We get some idea from a plane wave

$$\psi = Ae^{i(kx-\omega t)}$$

We know it has a fixed $k = \dfrac{2\pi}{\lambda}$, which means a fixed momentum

$$p = \frac{h}{\lambda} = \hbar k.$$

What operation on $Ae^{i(kx-\omega t)}$ gives $\hbar k$? Try $-i\hbar\dfrac{\partial}{\partial x}$.

$$-i\hbar\frac{\partial}{\partial x}\psi = -i\hbar\frac{\partial}{\partial x}Ae^{i(kx-\omega t)}$$

$$= -i\hbar\, ik\, Ae^{i(kx-\omega t)} = \hbar k\psi$$

Therefore, we identify the "operator" for momentum as

$$\boxed{\hat{p} = -i\hbar\frac{\partial}{\partial x}}$$

Average momentum is then given by

$$\langle \hat{p} \rangle = \int_{-\infty}^{\infty} dx\, \psi^*(x,t)\, \hat{p}\, \psi(x,t)$$

$$\langle \hat{p} \rangle = \int_{-\infty}^{\infty} dx\, \psi^*(x,t) \left(-i\hbar \frac{\partial}{\partial x} \right) \psi(x,t)$$

What is $\hat{p}^2$?

$$\hat{p}^2 = \left(-i\hbar \frac{\partial}{\partial x} \right) \left(-i\hbar \frac{\partial}{\partial x} \right) = -\hbar^2 \frac{\partial^2}{\partial x^2}$$

Same method applied using $i\hbar \frac{\partial}{\partial t}$ on the plane wave shows that:

$$i\hbar \frac{\partial}{\partial t} \psi(x,t) = E\psi(x,t)$$

So the operator for energy is

$$\boxed{\hat{E} = i\hbar \frac{\partial}{\partial t}}$$

Average energy is given by

$$\langle \hat{E} \rangle = \int_{-\infty}^{\infty} dx\, \psi^*(x,t)\, i\hbar \frac{\partial}{\partial t} \psi(x,t)$$

Once we find $\psi(x,t)$, we will try to do these integrals and understand how to practically do it.

Now we move on to the central question that we have been avoiding so far. Given a system, how to find $\psi(x,t)$?

Well, here is the answer: $\psi(x,t)$ is the solution of the following differential equation (called Schrödinger equation):

$$i\hbar \frac{\partial}{\partial t} \psi(x,t) = -\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} \psi(x,t) + V(x,t)\psi(x,t)$$

$V(x,t)$ specifies the details of the system, just like force was specified to us to use $F = ma$ in the old days.

See that this is the operator version of

$$E = \frac{p^2}{2m} + V(x,t)$$

acting on $\psi(x,t)$ from the left. Okay! That's the answer. Now we gotta solve it for given $V(x,t)$.

Usually we deal with potentials that don't change with time:

$$V(x,t) \longrightarrow V(x)$$

Then Schrödinger equation can be simplified.

Assume

$$\psi(x,t) = \phi(x)T(t)$$

Plugging this into Schrödinger equation,

$$i\hbar\frac{\partial}{\partial t}[\phi(x)T(t)] = -\frac{\hbar^2}{2m}T(t)\frac{\partial^2\phi(x)}{\partial x^2} + V(x)\phi(x)T(t)$$

Divide by ϕT:

$$\frac{i\hbar}{T}\frac{\partial T}{\partial t} = \frac{1}{\phi}\left[-\frac{\hbar^2}{2m}\frac{1}{\phi(x)}\frac{\partial^2\phi(x)}{\partial x^2}\phi(x) + V(x)\phi(x)\right]$$

LHS depends only on t.

RHS depends only on x.

So both should be equal to some constant, call it E.

$$i\hbar\frac{\partial T(t)}{\partial t} = ET(t) \tag{1}$$

$$-\frac{\hbar^2}{2m}\frac{\partial^2\phi(x)}{\partial x^2} + V(x)\phi(x) = E\phi(x) \tag{2}$$

This combination will appear often:

$$\underbrace{-\frac{\hbar^2}{2m}\frac{\partial^2}{\partial x^2} + V(x)}_{\hat{H}}\phi(x) = E\phi(x)$$

$$\boxed{\hat{H}\phi(x) = E\phi(x)}$$

$\hat{H}$ is called Hamiltonian or the total energy operator.

So, instead of one big Schrödinger equation, which is one second-order PDE in 2 variables, we have 2 ODEs, each of 1 variable.

Eq. (1) and (2) tell us that E is the total energy.

$$i\hbar\frac{\partial T(t)}{\partial t} = ET(t)$$

$$\Rightarrow i\hbar\frac{\partial T(t)}{\partial t}\phi(x) = ET(t)\phi(x)$$

$$\Rightarrow i\hbar\frac{\partial \psi(x,t)}{\partial t} = E\psi(x,t)$$

Solve Eq. (1). That's easy:

$$i\hbar\frac{\partial T(t)}{\partial t} = ET(t)$$

$$T(t) = e^{-iEt/\hbar}$$

Check!

That means

$$\psi(x,t) = \phi(x)e^{-iEt/\hbar}$$

All that is left is to solve

$$-\frac{\hbar^2}{2m}\frac{\partial^2\phi(x)}{\partial x^2} + V(x)\phi(x) = E\phi(x)$$

and get $\phi(x)$ and E.

These states have a property:

$$\psi(x,t) = \phi(x)e^{-iEt/\hbar}$$

$$\psi^*(x,t) = \phi^*(x)e^{iEt/\hbar}$$

$$|\psi(x,t)|^2 = \psi(x,t)\psi^*(x,t) = \phi(x)\phi^*(x)$$

$$= |\phi(x)|^2$$

Probability independent of time!

Called stationary states.